

♣ WILLIAM GILBERT, AND TERRES-
TRIAL MAGNETISM IN THE TIME OF
QUEEN ELIZABETH: A DISCOURSE
BY SILVANUS P. THOMPSON, F.R.S.

William Gilbert

❖ WILLIAM GILBERT, AND TERRES-
TRIAL MAGNETISM IN THE TIME OF
QUEEN ELIZABETH: A DISCOURSE
BY SILVANUS P. THOMPSON, F.R.S.

William Gilbert

The Project Gutenberg eBook of William Gilbert, and Terrestrial Magnetism in the Time of Queen Elizabeth

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: William Gilbert, and Terrestrial Magnetism in the Time of Queen Elizabeth

Author: Silvanus P. Thompson

Release date: June 5, 2014 [eBook #45893]

Most recently updated: October 24, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/45893

Credits: Produced by Chris Curnow, Charlie Howard, and the Online Distributed Proofreading Team at <http://www.pgdp.net> (This file was produced from images generously made available by The Internet Archive)

*** START OF THE PROJECT GUTENBERG EBOOK WILLIAM
GILBERT, AND TERRESTIAL MAGNETISM IN THE TIME OF QUEEN
ELIZABETH ***

 **WILLIAM GILBERT, AND
TERRESTRIAL
MAGNETISM IN THE
TIME OF
QUEEN ELIZABETH: A
DISCOURSE
BY SILVANUS P.
THOMPSON, F.R.S.**

WILLIAM GILBERT AND TERRESTRIAL MAGNETISM IN THE TIME OF QUEEN ELIZABETH.



WILLIAM GILBERT, the father of electrical science, was born in Colchester in 1540. Educated at St. John's College, Cambridge, where he took his degree as Doctor of Medicine in 1569, he settled, after four years of foreign travel, in London in 1573, and was admitted to the Royal College of Physicians, of which he became Censor, Treasurer, and, in 1599, President. He was in February, 1601, appointed personal Physician to the Queen, whom he attended in her last illness. He came of a well-known East Anglian family, and held extensive landed estates in Essex and Suffolk. He survived the Queen only eight months, dying November 30th, 1603.



GILBERT'S monumental work, the *De Magnete*, published in 1600, marks an era in magnetic science. For some four hundred years the employment of the magnetic needle in navigation had been known both in Northern and Southern Europe. While it is possible that the primitive use of the loadstone may be ascribed to the Baltic, it is certain that the employment of a pivotted needle, and the addition of a rose of the winds as a compass card both originated in the Mediterranean. The pivotted needle is described in the Epistle of Peter Peregrinus, written in 1269; while the earliest known compass-card marked

with the initials of the names of the winds is that ascribed to Jachobus Giraldis, of 1426, in the Biblioteca Marciana in Venice. The manner of use in Elizabethan times of the loadstone and of the compass may be gathered from Olaus Magnus (*Historia de Gentibus Septentrionalibus*, 1555), from Pedro de Medina (*Arte de Nauegar*, 1545), Martinus Cortes (*Breve compendio de la sphaera*, 1556), Blundevile (*Exercises*, 1594), Norman (*Newe Attractive*, 1581), Borough (*A discours of the Variation of the Cumpas*, 1581), Pedro Nuñez (*Instrumenta Artis Navigandi*, 1592), Barlow (*The Navigators Supply*, 1597), Nautonier (*Mécometrie de l'Eyman*, 1602), and Stevin (*Die Havenvinding*, 1599).

At the time when steering by the compass was introduced into navigation, the compass pointed in Middle Europe so nearly truly to the north that with the rough instrumental appliances at hand its deviation from the true north was seldom noticed, or if noticed ascribed to some error in the setting of the needle. Later the compass-makers began to set their needles slightly askew beneath the card, according to the variation in the place of origin. Norman (1581) states that those used in the Levant, made in Sicily, Genoa, or Venice, had the needles straight, while those used in Denmark and Flanders had them set at three-quarters of a point, or a whole point, to the eastward; while those made in Spain, Portugal, France, and England, had the needles set half a point to the east. Those for Russia were set at "three seconds of a point." Gilbert denounced these devices as tending to obscure the true facts. Gradually it became recognized, probably after the voyage of Columbus, when the manifest change in the declination of the needle nearly caused mutiny of the sailors, that the direction of the needle differs at different places; and accordingly navigators began to collect data. The record of the voyage of Columbus states that during his second voyage in 1496 he used for steering the observations made on the declination during his first voyage. The "secret" of Sebastian Cabot, which he declared when dying to be a divine revelation to him, can have been little else than the idea of using in navigation the local declinations of the compass. On the other hand, Pedro de Medina flatly denied the existence of the declination, adding that if the compass did not show the pole, the fault lay in the defective construction of the compass itself. Columbus had found a point $2\frac{1}{2}^{\circ}$ east of Corvo, in the Azores, where there was "no variation," and other navigators explored the "agonic" lines which crossed the Atlantic and the Indian ocean. According to Humboldt, Alonzo de Santa Cruz in 1530

constructed the first general variation chart. But along with this development of practical interest in the subject there grew up a crop of wild legends to account for the irregularities observed. The reason why the compass needle pointed north, and the reason why it did not point truly north, were alike proclaimed to be due to the stars, to the influence of spirits, or to the existence of loadstone mountains of uncertain locality and of fabulous power. The old traditions of the Arabian Nights, dressed in a newer setting, found themselves justified by the insertion in maps of loadstone rocks, the position of which changed at the fancy of the chartographer. Ptolemy had located them in the Manioles; Olaus Magnus declared them to be under the pole; Garzias ab Horto situated them in the region of Calcutta. The map of Johann Ruysch, which adorned the edition of Ptolemy, published at Rome in 1508, showed four magnetic islands in the Arctic Circle. Martinus Cortes placed the loadstone mountains in Sarmatia. Mercator in his great chart depicted two great rocks rising from the sea to the north of eastern Siberia, one being drawn on the supposition that at St. Michael the compass points due north, while the other is further north on the supposition that the compass points due north at Corvo. The map of Cornelius Wytfliet, 1597, shows the same phantom islands. Blundevile, writing in 1594 of the now lost map of Peter Plancius, mentions that he sets down the pole of the loadstone somewhat to southward of the islands that lie east of Groynelande.

Meantime another significant fact had been discovered in 1576 by Robert Norman, of Limehouse, compass-maker, namely, the tendency of the magnetized needle to dip its northern end downwards. Noticing this as a circumstance that occasioned him some trouble in the construction of his compasses, he thereupon devised a dipping-needle, and measured the dip, "which for this Cyty of London I finde by exact obseruations to be about 71 degrees 50 mynutes." He attributed both the declination and the dip of the needle to the existence of a "poynt respective," which the needle respected or indicated, but toward which it was not attracted. The first authoritative treatise on the variation of the compass was the tract by William Borough, comptroller to the Navy, who in 1580 found an eastward declination of $11^{\circ} 15'$ at Limehouse. Borough had himself travelled in northern regions and had found at Vaigats a westerly declination of 7 degrees, whereas by Norman's theory of the respective point there should have been an easterly declination of $49^{\circ} 22'$. The great navigators were continually bringing home

fresh information. Drake, Lynschoten, Cavendish, Hariot all contributed; as did lesser men such as Abraham Kendall, sailing-master to Sir Robert Dudley (the *soi-disant* Duke of Northumberland), and afterward companion of Drake in his last voyage. Teachers of navigation such as Simon Stevin of Bruges and Edward Wright, lecturer to the East India Company, might record and tabulate: but a master-mind was wanting to forge some larger and consistent doctrine which should afford a grasp of the whole subject. Such an one arose in Dr. William Gilbert. Nurtured, as we have seen, in the Cambridge which had so recently been the home of Linacre and of Kaye--the Kaye who founded Caius College--Gilbert had, during his subsequent sojourn in Italy, conversed with all the learned men of his time. He had experimented on the magnet with Fra Paolo Sarpi: he had, there is reason to think, met Giordano Bruno: he was the friend and correspondent of Giovanni Francesco Sagredo. Being a man of means and a bachelor, he spent money freely upon books, maps, instruments, minerals, and magnets. For twenty years he experimented ceaselessly, and read, and wrote and speculated, and tested his speculations by new experiments. For eighteen years he kept beside him the manuscript of his treatise, which in the year 1600 saw the light under the title of *De Magnete*, to which was added the sub-title: *magneticisque corporibus, et de magno magnete tellure, physiologia nova*. That which Gilbert had in fact perceived, and which none before him had glimpsed even dimly, was that the globe of the earth itself acted as a great loadstone, and that the tendency of the needle to point in a polar direction was due to the globe acting as a whole. So he boldly put into his title-page the statement that his new philosophy was concerning the great magnet the earth: and in chapter after chapter he set himself to describe the experiments upon which he founded his famous induction. The phrase *terrestrial magnetism* does not occur in any of the prior treatises, because the idea had not presented itself. Gilbert piled proof upon proof, sometimes most cogently, as when he constructed loadstone globes, or *terrellas* to serve as magnetic models of the earth; sometimes with indifferent logic, as when he pointed to the iron ore in the earth and reasoned that the magnet tended to conform to (*i.e.* turn itself toward) the homogenic substance of the body from which it had been dug. The local deviations of the compass he sought to account for by the irregularities of the earth's crust, and maintained that the compass tended always, at places off the coast of a continent, to be deflected somewhat toward that continent.

His syllogism was based on the fact that at that date all the way up the Atlantic seaboard of Europe, from Morocco to Norway, the variation was eastward. He argued that this was a universal law. But even within one generation, as may be seen in *Purchas his Pilgrims*, in the narrative of the voyage of Bylot and Baffin, the generality of the law was questioned. Gilbert reasoned on such knowledge as he had, and this did not include any notion of the secular changes in the declination. In his time, as he tells us, the variation of the compass at London was 11-1/3 degrees. What he did not know was that this was a diminishing quantity which in fifty-seven years would be reduced to zero, to be succeeded by a westward declination that would last for nearly three hundred years. For the facts as known in the thirty years succeeding Gilbert's death, see the remarkable and scarce volume of Gellibrand: *A Discourse Mathematical on the Variation of the Magneticall Needle* (1635).

Gilbert's treatise is a skilful literary achievement in which there is no trace to reveal whether any part was written before the rest. It is divided systematically into six books. The sixth book only appears to suffer from some incompleteness. It relates not so much to the magnet as to the Copernican theory of the universe, which doctrine Gilbert had eagerly espoused, and which he was the first in England to proclaim. It is known from a letter to Barlow, printed in 1616, that he intended to add to it certain chapters descriptive of some of his instruments, but he had not completed these before his death. The first book treats of historic accounts of the loadstone, of its origin and properties, of iron ores in general, and of the fables and vain opinions which in the handling of Paracelsus and of the schoolmen had grown up around the magnet. The second book is on the magnetic motions, and primarily on the attractions and repulsions between loadstones, between loadstone and iron, and between magnetic needles. In this book occurs the notable digression upon the subject of amber and the electric forces of amber and of other substances which when rubbed show, as he discovered, similar electrical powers. An analysis of this part, and a summary of Gilbert's electrical discoveries will be found in the Notes printed for the Gilbert Club to accompany the English translation (1900) of the *De Magnete*. After this digression Gilbert returns to the attractive properties of the loadstone, and to the way they are affected by giving it different shapes. In the course of this enquiry, he announces his discovery of the augmentation of the power of the loadstone by arming it with iron

caps, an invention which caused Galileo to say: "I extremely praise, admire, and envy this author for that a conception so stupendous should come into his mind. I think him moreover worthy of extraordinary applause for the many new and true observations that he has made." Gilbert further pointed out that the loadstone is surrounded by a sort of atmosphere or "orbe of virtue" within which the magnetical effects can be observed. Book 3, on the directive force of the magnet, is full of most instructive experiments, in which the terrella figures largely, relating to the question how one magnet influences another and tends to make it point toward it. All this was leading up to the theory of terrestrial magnetism; for we find him naming the parts of his loadstone globes with poles, equator and meridians. In this book he dilates on the observation that vertical iron rods, such as the finial on the Church of St. John at Rimini, spontaneously acquired magnetic properties. This he traced to the influence of the earth, and demonstrated the effect by magnetizing iron bars by simply hammering them on the anvil while they lay in a north and south position. Book 4 deals with the Declination, or, as it was then called, the variation of the compass. He discusses its observation and measurement, the influence of islands, the results obtained by travellers to distant parts, Nova Zembla, the Guinea coast, the Canary Isles, Florida, Virginia, Cape Race, and Brazil. Then he recounts his experiments with terrellas having uneven surfaces to represent the irregularities of the earth's crust. He points out errors arising from the fallacious practice of setting the needle obliquely under the card. He considers in separate chapters the variations in Nova Zembla, in the Pacific, in the Mediterranean, and in the Eastern Ocean. The fifth book is on the Dip. Gilbert seized with avidity on Norman's discovery of this effect, and devised an improved form of dipping-needle. He experimented on the dip of compass-needles placed at different points over his terrella, and evolved a theory on the proportion which he conceived to exist between the latitude and the dip. Arguing from all too imperfect data, he propounded the view that the dip was the same in any given latitude; and proposed that seamen should ascertain their latitudes by simply observing the dip. He was aware that local irregularities might occur, as they do in the declination; but was not deterred by this knowledge from propounding his theory with much circumstance and considerable geometrical skill. After the publication of his book he developed the theory still further and gave it to Blundevile for publication. At Gilbert's suggestion Briggs of Gresham College calculated out a table of dip and

latitude. It was, however, soon found that the facts deviated more or less widely from the theory. Further observations in other lands showed the method to be impracticable; and Gilbert's hope to give to the mariner a magnetic measure of latitude remained unfulfilled. Book 5 closes with an eloquent passage in which Gilbert affirmed the neo-Platonic doctrine of the animate nature of the universe, and asserted that Thales was right when he held (as Aristotle relates in the *De Anima*) that the loadstone was animate, being part of and indeed the choice offspring of its animate mother the earth. Book 6, as already mentioned, is devoted to Copernican ideas, and contains Gilbert's one contribution to the science of Astronomy, in his remark that the fixed stars (previously regarded as fixed in the eighth of the celestial spheres at one common distance from the central earth) were in reality set in the heavens at various distances from the earth.

From this brief analysis it will be seen that Gilbert's claims to eminence rest not upon any particular discovery or invention, but upon his having built up a whole experimental magnetic philosophy on a truly scientific basis, in place of the vague and wild speculations which had previously been accepted. By his magnificent generalization from the small scale models to the globe itself, supported from point to point by experimental researches, he created the science of terrestrial magnetism. If from the imperfection of the data at his disposal he fell into sundry errors of detail, he yet founded the method of philosophizing by which those errors were in due time corrected. And if for nothing else than his masterly vindication of scientific method, and his rescue of the subject of magnetism from the pedantry and charlatanry into which in the preceding ages it had lapsed, his memory must be held in high honour.

Alas that of the personality of so great a man so little should be known. A brief but characteristic biography of him is enshrined by old Fuller in his *Worthies*. The poet Dryden, and the epigrammatist Owen, celebrated him in still briefer verse. His portrait, which hung for nigh two hundred years in the Schools Gallery, at Oxford, disappeared a century ago, leaving only a poor engraving to perpetuate his scholarly countenance. Doubtless he is one of the four physicians depicted by the pencil of Camden in his famous cartoon (now in the British Museum), as walking in the funeral procession of Queen Elizabeth. Of his handwriting not a vestige was known until about five years ago, when a signature was unearthed in the Record Office.

Subsequently four signatures were found in the books of St. John's College; and recently there has come to light a volume of Aristotle bearing Gilbert's own marginal notes. His will lies at Somerset House, but it is only a copy. Of his fine collection of minerals and loadstones, which with his maps, books, manuscripts, and correspondence with Sarpi and Sagredo and others, he bequeathed to the College of Physicians, nothing remains: they perisht in the Great Fire of London. In a quiet corner of the City of Colchester stands the quaint old house where he lived, and where, according to local tradition, he once received the Queen. And hard by it is the church of Holy Trinity, in which a mural tablet records his virtues and marks his last resting place. But his true monument is the immortal treatise in which he laid the foundations of terrestrial magnetism and of the experimental science of electricity.

To the names of the men who made great the age of Queen Elizabeth, who added lustre to the England over which she ruled, and made it famous in foreign discovery, in sea-craft, in literature, in poetry, and in drama, must be joined that of the man who equally added lustre in science, Doctor William Gilbert.

This discourse on William Gilbert and terrestrial magnetism
in the time of Queen Elizabeth was delivered by Silvanus
P. Thompson at the meeting of the Royal Geographical
Society on March twenty-third MDCCCCIII on the
occasion of the tercentenary of the death of
Queen Elizabeth, and is now printed
by Charles Whittingham
and Company at the
Chiswick Press.



Transcriber's note: The original book used the long "s", which has been changed to the modern "s" here.

Other archaic spellings have not been changed.

*** END OF THE PROJECT GUTENBERG EBOOK WILLIAM
GILBERT, AND TERRESTIAL MAGNETISM IN THE TIME OF QUEEN
ELIZABETH ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.